

1D CNN — Architecture Variant Comparison

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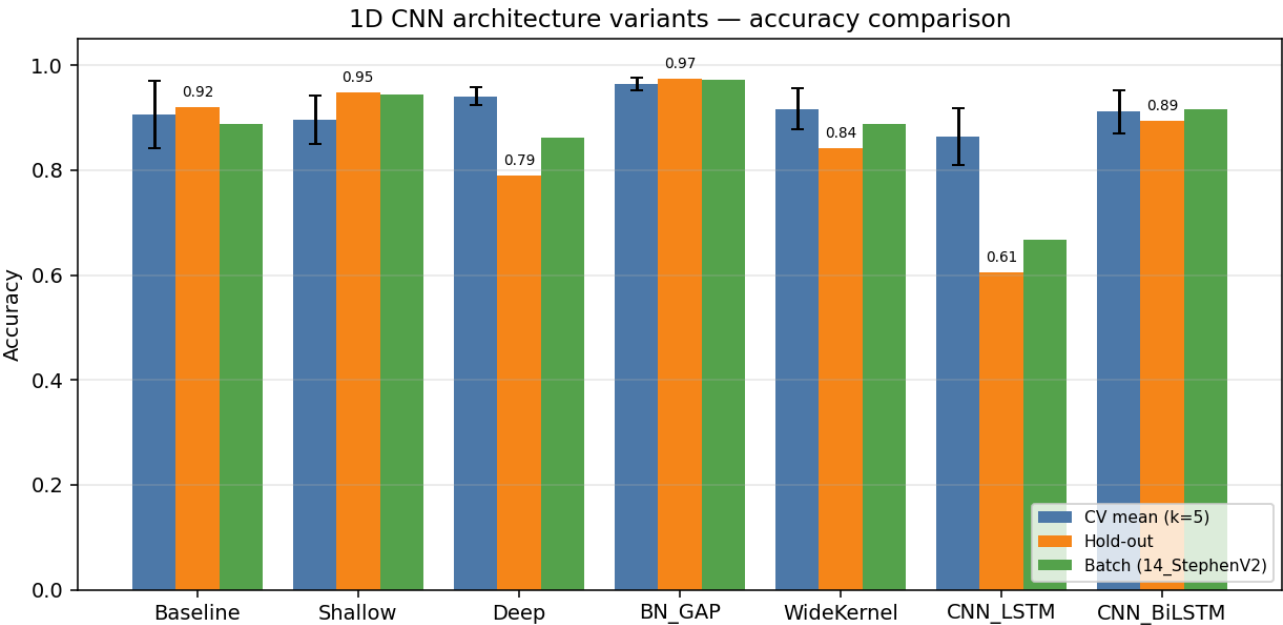
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Total trials	376 (train pool 338, hold-out 38)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

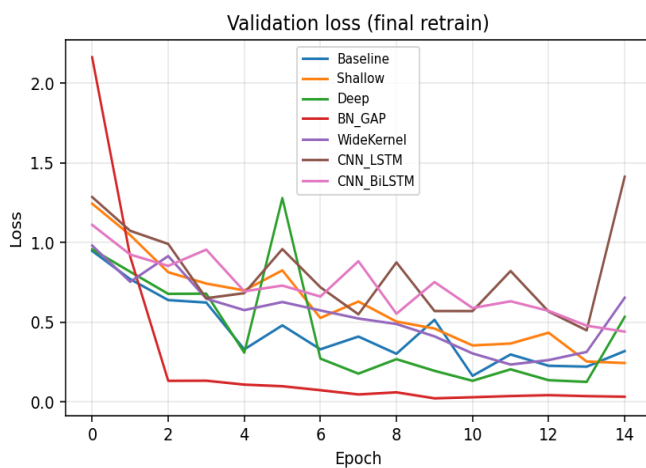
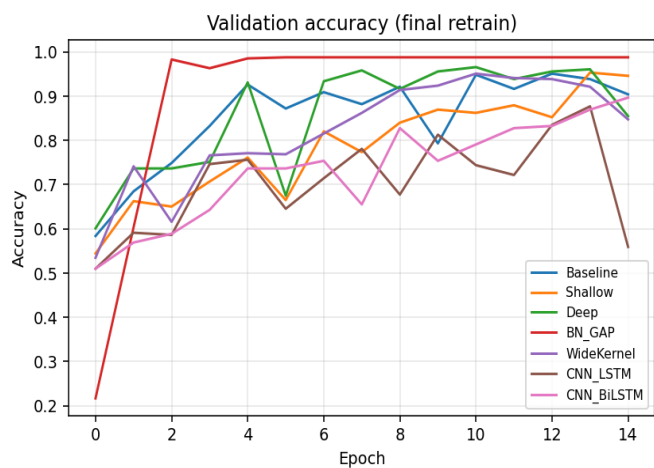
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9056	0.0640	0.9211	0.2746	0.8889	32/36
Shallow	72,487	0.8967	0.0462	0.9474	0.1324	0.9444	34/36
Deep	204,007	0.9407	0.0166	0.7895	0.5758	0.8611	31/36
BN_GAP	57,319	0.9645	0.0117	0.9737	0.1188	0.9722	35/36
WideKernel	131,687	0.9169	0.0387	0.8421	0.7509	0.8889	32/36
CNN_LSTM	68,455	0.8640	0.0544	0.6053	1.1376	0.6667	24/36
CNN_BiLSTM	105,575	0.9112	0.0413	0.8947	0.2861	0.9167	33/36

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9056 ± 0.0640
Hold-out test acc / loss	0.9211 · 0.2746
Batch-test acc	0.8889 (32/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9265	0.9706	0.9265
2	1620	68	0.7794	0.9559	0.7794
3	1620	68	0.9265	0.9412	0.9265
4	1626	67	0.9403	0.9403	0.9403
5	1626	67	0.9552	0.9701	0.9552

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.800	0.889	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.946	0.914	0.915	38
weighted avg	0.951	0.921	0.922	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	6	0.333
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	36	0.889

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8967 ± 0.0462
Hold-out test acc / loss	0.9474 · 0.1324
Batch-test acc	0.9444 (34/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.8088	0.8235	0.8088
2	1620	68	0.9118	0.9265	0.9118
3	1620	68	0.8971	0.8971	0.8971
4	1626	67	0.9254	0.9254	0.9254
5	1626	67	0.9403	0.9403	0.9403

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.959	0.943	0.940	38
weighted avg	0.962	0.947	0.945	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	6	0.667
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	36	0.944

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9407 ± 0.0166
Hold-out test acc / loss	0.7895 · 0.5758
Batch-test acc	0.8611 (31/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9559	0.9706	0.9559
2	1620	68	0.9412	0.9853	0.9412
3	1620	68	0.9559	0.9559	0.9559
4	1626	67	0.9104	0.9104	0.9104
5	1626	67	0.9403	0.9701	0.9403

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	1.000	0.200	0.333	5
Double_Pistol_Recoil	1.000	0.667	0.800	6
Double_Raise	0.667	1.000	0.800	6
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	0.833	0.909	6
Drum_Roll	0.667	0.800	0.727	5
macro avg	0.840	0.786	0.759	38
weighted avg	0.844	0.789	0.765	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	6	0.667
Double_Pistol_Recoil	2	5	0.400
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	31	36	0.861

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9645 ± 0.0117
Hold-out test acc / loss	0.9737 · 0.1188
Batch-test acc	0.9722 (35/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9412	0.9853	0.9412
2	1620	68	0.9706	0.9706	0.9706
3	1620	68	0.9706	0.9853	0.9706
4	1626	67	0.9701	0.9851	0.9701
5	1626	67	0.9701	0.9701	0.9701

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	38
weighted avg	0.978	0.974	0.973	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	6	0.833
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	35	36	0.972

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9169 ± 0.0387
Hold-out test acc / loss	0.8421 · 0.7509
Batch-test acc	0.8889 (32/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9265	0.9265	0.9265
2	1620	68	0.9706	0.9706	0.9706
3	1620	68	0.9412	0.9412	0.9412
4	1626	67	0.8806	0.9254	0.8806
5	1626	67	0.8657	0.9701	0.8657

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.600	0.750	5
Double_Nothing	0.750	0.600	0.667	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	0.667	0.800	6
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	0.857	1.000	0.923	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.880	0.838	0.836	38
weighted avg	0.886	0.842	0.842	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	6	0.833
Double_Pistol_Recoil	5	5	1.000
Double_Raise	2	5	0.400
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	36	0.889

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8640 ± 0.0544
Hold-out test acc / loss	0.6053 · 1.1376
Batch-test acc	0.6667 (24/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9265	0.9265	0.9265
2	1620	68	0.7647	0.9265	0.7647
3	1620	68	0.8676	0.8676	0.8676
4	1626	67	0.8955	0.9254	0.8955
5	1626	67	0.8657	0.9403	0.8657

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.625	1.000	0.769	5
Double_Nothing	0.333	0.600	0.429	5
Double_Pistol_Recoil	0.000	0.000	0.000	6
Double_Raise	0.667	1.000	0.800	6
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	0.500	0.667	6
Drum_Roll	1.000	0.200	0.333	5
macro avg	0.607	0.614	0.538	38
weighted avg	0.603	0.605	0.534	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	6	0.833
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	24	36	0.667

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.9112 ± 0.0413
Hold-out test acc / loss	0.8947 · 0.2861
Batch-test acc	0.9167 (33/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9118	0.9265	0.9118
2	1620	68	0.9412	0.9853	0.9412
3	1620	68	0.9118	0.9265	0.9118
4	1626	67	0.8358	0.9403	0.8358
5	1626	67	0.9552	0.9701	0.9552

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	0.600	0.750	5
macro avg	0.937	0.886	0.888	38
weighted avg	0.942	0.895	0.897	38

Batch test per class (14_StephenV2)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	6	0.667
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	33	36	0.917